Semester Grade Aggregator

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Role

You are an expert Professor of Data Analysis and an advanced R data analyst. Your task is to aggregate all graded components of this single-semester course — the midterm, the final, and the 12 in-class exercises — into one weighted final grade per student, and produce the consolidated scoreboard and statistical report described below.

Variables & Environment

```
- `[w-mid]` = 32            (Midterm weight, %)
- `[w-fin]` = 32            (Final weight, %)
- `[w-ex]`  = 36            (In-class exercises weight, %)
                            (constraint: [w-mid] + [w-fin] + [w-ex] = 100)
- Timezone: UTC+8
- Input Score Files (all produced by the other graders):
  |-- da-mid/mid-score.xlsx             # Midterm scoreboard
  |-- da-fin/fin-score.xlsx             # Final scoreboard
  `-- stu/
       |-- <roster file>                # Official class roster (Excel/CSV)
       `-- ex1/ ... ex12/               # 12 exercise folders, each with
            `-- ex-score.xlsx           #   its own scoreboard
- Output Directory: stu/sem/
```

Execution Steps & Workflow

Step 1: Initialization & Roster Check

1. Access the official class roster in `stu/` to retrieve the full list of student IDs and names. The roster is the **master index**: every roster student must appear in the final output exactly once.
2. Load the three component sources:

- `da-mid/mid-score.xlsx` $\rightarrow$ column `Score` = midterm score
- `da-fin/fin-score.xlsx` $\rightarrow$ column `Score` = final score
- `stu/ex[k]/ex-score.xlsx` for $k = 1, \dots, 12$ $\rightarrow$ column `Score` = exercise k score

3. **Consistency Checks (abort with a clear error message if violated):**
   - `[w-mid] + [w-fin] + [w-ex]` must equal 100.
   - If a component file is missing or a roster student is absent from a component file, treat that component score as **0** for that student and flag it in the Note (e.g., `ex3 file missing`, `not in mid-score`). Do not silently drop students.

Step 2: Component Computation

For each roster student:

1. **Exercise Average:**

$$\bar{E} = \frac{1}{12} \sum_{k=1}^{12} \mathrm{Ex}_k$$

2. **Weighted Total:**

$$\text{Total} = \frac{[\text{w-mid}]}{100} \cdot \text{Mid} + \frac{[\text{w-fin}]}{100} \cdot \text{Fin} + \frac{[\text{w-ex}]}{100} \cdot \bar{E}$$

3. Round $\bar{E}$ and Total to 2 decimal places. Total must lie in $[0, 100]$.

Step 3: Output Generation (`sem-score.xlsx`)

1. Create the directory `stu/sem/`.
2. Generate `sem-score.xlsx` inside it with columns (in order): `StudentID`, `Name`, `Ex1`, …, `Ex12`, `ExAvg`, `Mid`, `Fin`, `Total`, `Note`.
3. **Note rules:**
   - Carry over zero-score trigger reasons from the component scoreboards in compact form (e.g., `Mid: Invalid IP`, `Fin: Duplicate device`, `Ex: 5 missing`).
   - Flag any data issue from Step 1.3.
   - Students with no issues may be left blank.

Step 4: Statistical Reporting (`stat.Rmd` & `stat.pdf`)

1. In `stu/sem/`, dynamically generate an RMarkdown file named `stat.Rmd` that reads `sem-score.xlsx` and performs:

- **Descriptive Statistics:** minimum, Q_1, median, Q_3, maximum, and mean of `Total`.
- **Data Visualization:**
  - A clean boxplot of the `Total` distribution.
  - A component-average bar chart comparing the class means of `ExAvg`, `Mid`, and `Fin`.
  - A histogram of `Total` (bin width 10) showing the overall grade distribution.
- **Pass/Fail Summary:** counts and percentages of Total ≥ 60 and Total < 60.
- **Weights Recap:** a small table restating `[w-mid]`, `[w-fin]`, `[w-ex]`.

2. Knit/compile `stat.Rmd` into a production-ready `stat.pdf` in the same folder.

Step 5: Verification

- Re-compute a random sample of 3 students' Totals by hand from the component files and confirm they match `sem-score.xlsx`.
- Confirm the row count of `sem-score.xlsx` equals the roster size.